GOS Case Study Report

Group E

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November 12, 2026

1 Introduction

In this case study, we assume the role of a consulting team tasked with guiding a Swiss pharmaceutical company through a period of significant uncertainty. Our objective is to analyse the current challenges facing the industry, assess their potential impact on the company's operations and profitability, and develop a comprehensive strategy that mitigates risks and leverages emerging opportunities.

Among the most pressing challenges are the introduction of new taxes, the ongoing discussion surrounding the implementation of Most-Favored-Nation clauses, and broader geopolitical tensions that disrupt supply chains and restrict market access. Together, these developments contribute to an increasingly unpredictable economic environment that requires strategic foresight, flexibility, and careful risk management.

The pharmaceutical industry is also a key pillar of the Swiss economy and the main contributor to Switzerland's GDP growth. Consequently, it is crucial to understand how companies within this sector can effectively manage uncertainty and maintain competitiveness. A significant number of jobs and livelihoods depend on the continued strength and stability of the pharmaceutical industry. Currently, roughly three quarters of Swiss pharmaceutical exports face considerable uncertainty, highlighting the urgent need for strategies that help firms prepare for and navigate these challenges.

To address these issues, this case study examines the broader political risks to which the pharmaceutical industry is exposed and analyses the degree to which Switzerland's pharmaceutical sector depends on the United States as a key market and partner.

In addition, we look at several strategic options available to the company. In particular, we analyze the potential relocation of production as well as the possibility of a strategic market exit. These strategic alternatives are supported by a financial assessment in order to determine the most sustainable solution for the company. (Interpharma 2025)

2 Swiss pharmaceutical industry landscape

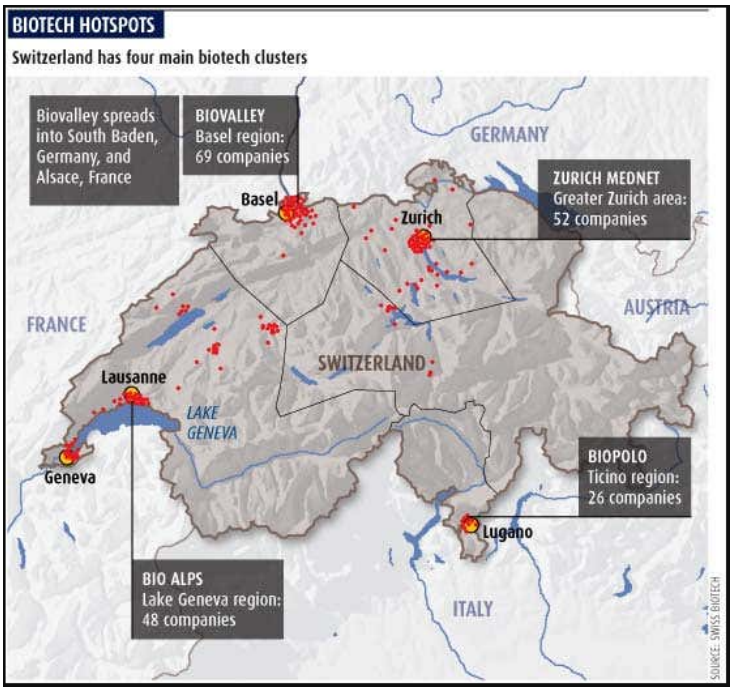


Figure 1: Overview Pharmaceutical industry Hubs in Switzerland

Switzerland's pharmaceutical industry is a central pillar of the national economy, with exports reaching CHF 114.5 billion and accounting for 40.5% of total exports. The sector contributes 5.8% of GDP directly and 9.8% including indirect effects (CHF 74.5 billion), while supporting 50,600 direct jobs and 250,200 additional jobs.The industry is organized around several complementary clusters, mainly located in Basel, Zurich, the Lake Geneva region, and Ticino. Basel represents the core pharmaceutical hub, hosting Roche and Novartis. In 2025, Roche generated approximately USD 77.5 billion (CHF 69.75 billion) in revenue, while Novartis reached USD 50.3 billion (CHF 45.27 billion). Together, these two firms represent roughly 85% of Switzerland's pharmaceutical export value, explaining the strong concentration of the industry in Basel.

They are complemented by Lonza, a leading CDMO (Contract Development and Manufacturing Organization), with CHF 6.53 billion in revenue in 2025. This reflects a vertical specialization within the Swiss ecosystem, combining innovation-driven pharmaceutical companies with manufacturing specialists. In addition, international firms such as Johnson & Johnson, Pfizer, and Moderna have established operations in Switzerland, attracted by its strong research environment and high-value activities. Switzerland's attractiveness is strongly linked to its scientific ecosystem. Institutions such as ETH Zurich and EPFL are consistently ranked among the world's top universities, particularly in engineering, biotechnology, and life sciences. This provides pharmaceutical companies with access to highly skilled talent and cutting-edge research. Switzerland also ranks first globally in talent competitiveness and staff training according to the Global Talent Competitiveness Index and the World Economic Forum. In addition, the country offers a high quality of life, which helps attract and retain international professionals.

Another key advantage is Switzerland's institutional position relative to the European Union. Although Switzerland is not part of the EU or the EEA, it maintains strong links through bilateral agreements. In particular, the Mutual Recognition Agreement (MRA) on Good Manufacturing Practices (GMP) allows Swiss pharmaceutical products to meet EU quality standards without additional certification. Switzerland's regulatory agency, Swissmedic, also cooperates closely with the European Medicines Agency (EMA), facilitating smoother regulatory processes while maintaining strong ties with the US FDA. This gives pharmaceutical firms privileged access to both the European and US markets. These

structural advantages translate into strong economic performance. The pharmaceutical sector has grown by 14.5% per year over the last decade and has driven nearly half of Switzerland's GDP growth. However, recent trends point to increasing pressure, including a CHF 560 billion shortfall in foreign investment between 2018 and 2023 and rising geopolitical and regulatory uncertainty.

Overall, Switzerland's pharmaceutical industry combines a highly concentrated core of global leaders with a diversified and innovation-driven ecosystem, explaining its strong international position despite growing long-term challenges.

3 Global footprint and US exposure

3.1 Analysis of the operational and economic reliance

Roche was selected as a case study due to its strong global footprint, its listing on the Swiss stock exchange, and its significant presence in the United States, making it particularly relevant for analyzing both operational and economic reliance on a single market. In this context, economic reliance can be formally defined as the degree to which a firm depends on a specific country for its revenues, profitability, and overall financial performance, while operational reliance refers to the extent to which a firm depends on a country for core activities such as production, research and development (R&D), supply chain, and workforce. Roche demonstrates a very high level of economic reliance on the United States. According to the 2025 Roche Annual Report, the Pharmaceuticals Division generated 53.2% of its sales in the US in 2025, while at the Group level, the United States accounted for approximately CHF 29.5 billion out of CHF 61.5 billion total sales (≈48%), with +8% growth between 2024 and 2025, confirming that the US is both the largest market and the main growth engine. This reflects a structural reliance (a long-term dependence embedded in the business model rather than temporary exposure).

Multi-year overview

Statistics, as reported

	2021	2022	2023	2024	2025
Income statement in millions of CHF					
Sales	62,801	63,281	58,716	60,495	61,516
EBITDA	24,692	25,015	21,976	23,538	24,608
Operating profit	18,155	17,476	15,395	13,417	18,476
Net income attributable to Roche shareholders	13,930	12,421	11,498	8,277	12,880
Research and development	14,799	15,225	14,200	15,304	13,352
Balance sheet in millions of CHF					
Non-current assets	56,690	54,335	57,022	61,765	61,974
Current assets	35,627	33,816	33,446	40,036	38,729
Total assets	92,317	88,151	90,468	101,801	100,703
Non-current liabilities	(25,556)	(28,897)	(32,381)	(38,617)	(34,822)
Current liabilities	(38,416)	(27,239)	(24,824)	(27,023)	(28,001)
Total liabilities	(63,972)	(56,136)	(57,205)	(65,640)	(62,823)
Net assets	28,345	32,015	33,263	36,161	37,880
Capital and reserves attributable to Roche shareholders	24,489	27,992	29,315	31,767	33,802
Equity attributable to non-controlling interests	3,856	4,023	3,948	4,394	4,078
Additions to property, plant and equipment	3,826	3,402	3,770	3,495	3,619
Key ratios					
Net income attributable to Roche shareholders as % of sales	22	20	20	14	21
Net income attributable to Roche shareholders as % of equity	57	44	39	26	38
Research and development as % of sales	24	24	24	25	22
Current ratio %	93	124	135	148	138
Equity and non-controlling interests as % of total assets	31	36	37	36	38
Data on shares and non-voting equity securities					
Number of shares	160,000,000	106,691,000	106,691,000	106,691,000	106,691,000
Number of non-voting equity securities (*Genussscheine*)	702,562,700	702,562,700	702,562,700	702,562,700	702,562,700
Total shares and non-voting equity securities	862,562,700	809,253,700	809,253,700	809,253,700	809,253,700
Total dividend in millions of CHF	7,526	7,688	7,769	7,850	7,931[a]
Earnings per share and non-voting equity security (diluted) in CHF	16.20	15.37	14.31	10.31	16.04
Dividend per share and non-voting equity security in CHF	9.30	9.50	9.60	9.70	9.80[a]

Information in this table is stated as reported and changes in accounting policies arising from changes in IFRS Accounting Standards are not applied retrospectively.

a) 2025 dividend proposed by the Board of Directors.

Figure 2: *Roche Group sales by geographical area (2021–2025). Source: Roche Annual Report 2025, p. 170.*

This dependence is reinforced by product-level concentration. Ocrevus (a treatment for multiple sclerosis, a chronic disease in which the immune system attacks nerve fibers) generated CHF 7.01 billion in 2025 compared to CHF 6.74 billion in 2024 (+9%), with 69.5% of its sales in the US. Xolair (a treatment for

asthma and allergic diseases) generated CHF 3.1 billion in 2025, growing by +32% between 2024 and 2025, while Vabysmo (used to treat macular degeneration, a disease affecting central vision) reached CHF 4.1 billion (+12%), with around 70% of sales in the US. These figures show that Roche's performance is strongly linked to the US both geographically and through its most important products.
This leads to a dual dependency structure, as Roche depends both on a single market and on a limited number of products. The top five drugs generated CHF 21.4 billion in 2025, representing nearly 45% of pharmaceutical sales, and increased by CHF 3.2 billion compared to 2024, confirming strong revenue concentration.

Hemlibra	4,754	4,503	+11	10.0	9.8
Polivy	1,470	1,121	+38	3.1	2.4
Gazyva/Gazyvaro	986	910	+14	2.1	2.0
MabThera/Rituxan[a]	962	1,116	-9	2.0	2.4
Columvi	288	172	+75	0.6	0.4
Others	160	87	+93	0.3	0.1
Total Haematology	***8,620***	***7,909***	***+15***	***18.1***	***17.1***
Total Oncology/Haematology	**23,938**	**23,744**	**+6**	**50.2**	**51.4**
Neurology					
Ocrevus	7,010	6,744	+9	14.7	14.6
Evrysdi	1,757	1,631	+13	3.7	3.5
Enspryng	364	311	+23	0.8	0.7
Madopar	350	368	0	0.7	0.8
Elevidys	341	189	+84	0.7	0.4
Others	12	24	-47	0.0	0.1
Total Neurology	**9,834**	**9,267**	**+11**	**20.6**	**20.1**
Immunology					
Xolair	3,075	2,470	+32	6.5	5.3
Actemra/RoActemra	2,470	2,645	-2	5.2	5.7
Pulmozyme	479	455	+12	1.0	1.0
CellCept	385	399	+1	0.8	0.9
MabThera/Rituxan[a]	289	263	+16	0.6	0.6
Others	34	97	-64	0.0	0.2
Total Immunology	**6,732**	**6,329**	**+12**	**14.1**	**13.7**
Ophthalmology					
Vabysmo	4,102	3,864	+12	8.6	8.4
Others	108	166	-31	0.2	0.3

Figure 3: *Main pharmaceutical growth drivers and product concentration in 2025. Source: Roche Annual Report 2025, pp. 15–20.*

Roche also shows strong operational reliance on the United States, with more than 25,000 employees, 13 manufacturing sites, and 15 R&D centers located in the country. This dependence is further reinforced by a major strategic commitment: in 2025, Roche announced a USD 50 billion investment plan over five years, expected to create more than 12,000 jobs and expand both research and production infrastructure across several US states. This reflects a strategic shift toward localising production and innovation in its most important market. In addition, the company depends on a highly concentrated distribution network, as three US distributors account for approximately one-third of total Group sales. Roche's model is also exposed to lifecycle risk, as revenues from older drugs decline after patent expiration. According to the 2025 report, Avastin, Herceptin, and MabThera declined from CHF 3.99 billion in 2024 to CHF 3.25 billion in 2025 (–14%), while Lucentis declined by approximately –58% in the US. This erosion is driven by biosimilars such as Truxima and Rixathon, which replicate original drugs at lower prices. This demonstrates increasing competition from global players, particularly from Asia, leading to price pressure and loss of market share.
Competition in the pharmaceutical industry is increasingly global and innovation-driven. Roche reported 51 new partnerships and 4 acquisitions in 2025 while maintaining a large innovation pipeline, with 66 molecules in development and 10 entering phase III clinical trials targeting diseases such as cancer and Alzheimer's. This confirms that competition is no longer regional but global, with new entrants

accelerating innovation cycles and pricing pressure. Financially, this model also increases pressure on resources. Roche's free cash flow decreased by 15% to CHF 11.8 billion in 2025 compared to 2024, reflecting higher investment needs and rising capital intensity. Finally, geographic diversification remains limited, as weaker performance in Asia, particularly in China, does not fully offset Roche's dependence on the US market. Overall, Roche's economic and operational reliance on the United States remains deep and structural.

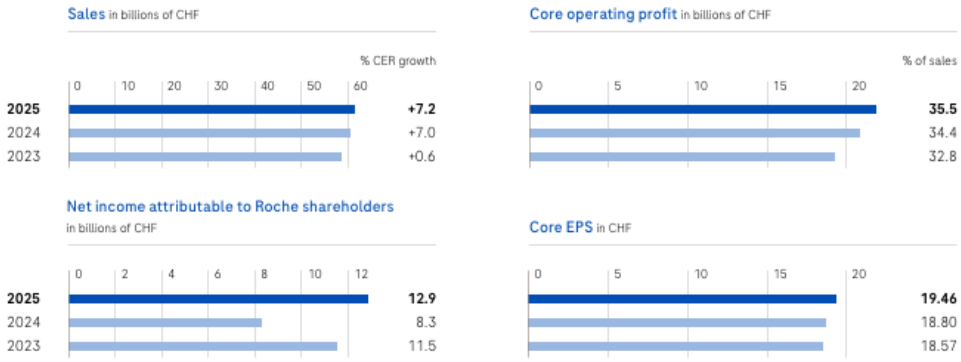


Figure 4: *Roche financial performance and free cash flow evolution. Source: Roche Annual Report 2025, pp. 2–5 and Financial Review section.*

3.2 Strengths and weaknesses of the global network at Roche

Roche's global network provides major advantages in terms of scale, innovation, and global impact. The company operates in more than 100 countries and employed 112,774 people worldwide in 2025, with approximately 46% in Europe, 35% in the Americas, and 19% in Asia-Pacific. This global presence provides access to diverse talent pools and supports complex scientific and industrial operations. Roche also demonstrates strong innovation capacity, with CHF 12.243 billion invested in R&D in 2025, representing around 20% of total sales. Its pipeline remains strong, with 66 molecules under development and 10 entering phase III clinical trials in 2025. The company's global impact is reflected by 39 million patients treated, 258 million people reached through diagnostics, and approximately 31 billion diagnostic tests delivered. Roche also launched 61 new diagnostic solutions in 2025, illustrating its ability to innovate at scale and integrate diagnostics with pharmaceuticals. In addition, Roche successfully translates innovation into commercial growth, with products such as Xolair and Polivy growing by 32% and 38% respectively between 2024 and 2025.

Despite these strengths, Roche's global network also presents several structural weaknesses. The most significant is its strong dependence on the United States, which represented approximately 48% of total Group sales in 2025. Revenue concentration is also high, as around 45% of pharmaceutical sales are generated by only five products, increasing vulnerability to product-specific shocks. The company is further exposed to distribution concentration, with approximately one-third of total sales depending on three US distributors.In addition, increasing competition from biosimilar producers, particularly in Asia, accelerates price erosion and reduces market share for older products. Financial pressure is another weakness, as Roche's free cash flow declined by 15% in 2025 due to rising R&D and investment costs. Internal imbalances also remain visible, with Pharmaceuticals significantly outperforming Diagnostics (49.2% margin versus 14.4% in 2025). Finally, weaker performance in Asia limits Roche's ability to reduce its dependence on the US market. Overall, Roche's global model combines strong innovation capabilities and global reach with significant strategic dependencies. While the company remains highly competitive and profitable, its long-term success depends on maintaining innovation while reducing exposure to market concentration and growing global competition.

4 Political Risk Evaluation

Roche generates approximately 48% of group revenues in the United States (Roche Holding AG 2025). This concentration exposes the company to political risks that go beyond normal market volatility. In this section three main risks were evaluated: the US tariffs on pharmaceutical imports, the Most Favored Nation (MFN) drug pricing executive order, and supply chain disruption coming from geopolitical instability. A ranked prioritization that connects to the strategic options analyzed in sections 5 and 6 is presented.

4.1 US tariff risk

The US government imposed 100% tariffs on pharmaceutical imports, justifying this as a national security measure under Section 232 of the Trade Expansion Act of 1962 (Reuters 2026). This unilateral executive action grants exemptions to companies that expand their US manufacturing or comply with government pricing agreements.
In 2025 the Pharmaceuticals Division of Roche generated 53.2% of its sales in the US equal to CHF 25.4 billion (Roche Holding AG 2025). Roche's main mitigation action is its announced USD 50 billion US investment plan, covering manufacturing and R&D expansion across several states over five years (Roche Holding AG 2025). This secured up to now three concessions: a tariff exemption lasting three years, a faster FDA approval pathway for new treatments, and the exclusion from Medicare price negotiations (Reuters 2026). Independent sources, like Public Eye (2025), claim however that the net financial damage from tariffs, given these concessions, is limited since Roche maintained a solid 31% net profit margin in 2024 throughout the policy escalation.
The residual risk depends now on the fulfilment of the plan, meaning that the tariff exemption holds only if Roche executes the investment commitment. Free cash flow fell 15% to CHF 11.8 billion in 2025 (Roche Holding AG 2025). If capital spending is deferred, the political protection becomes much weaker. The investment plan also assumes that current US political conditions hold through 2030.
The tariff risk is thus very relevant but not so likely to happen in full force also considering that the investment plan has largely neutralized the short-term exposure.

4.2 MFN drug pricing

The Trump administration's May 2025 executive order foresees the communication of MFN price targets to manufacturers, with effective translation as a fallback if the voluntary agreements will fail. This internal pricing directive, even if it shares the MFN name with the WTO principle, operates through a very different mechanism.
The December 2025 agreement applied discounts of 70% to 90% to a narrow group of drugs that are not bestsellers, targeting Medicaid patients who already receive large statutory rebates (Reuters 2026). These discounts were calculated against inflated list prices rather than actual net prices. According to Public Eye (2025) the real revenue impact can be considered negligible as Roche's core product revenues were largely unaffected. In exchange, Roche received the tariff exemption, faster approvals, and Medicare exclusion noted in section 4.1. The agreement is a first round settlement on a narrow, low revenue product set. Whether MFN pricing will be enforced on Roche's highest revenue biologics remains open.
Roche's exposure concentrates in three products, all of them treating chronic condition and not facing generic or alternative competition, which removes the price discipline that generic alternatives would otherwise provide.
Ocrevus is a treatment for multiple sclerosis which generated CHF 7.01 billion in 2025, with 69.5% of sales in the US (Roche Holding AG 2025). It is a single-source biologic with no approved alternatives or generic treatments. The US sales are approximately CHF 4.87 billion so a 20% US price reduction would lead to an annual revenue loss of approximately CHF 975 million from this product alone (Roche Holding AG 2025).
Vabysmo is an ophthalmic treatment for macular degeneration which reached CHF 4.1 billion in 2025,

with more than half of the sales in the US (roughly 70%) (Roche Holding AG 2025). The US revenue is approximately CHF 2.87 billion thus a 20% reduction translates into CHF 574 million in annual impact (Roche Holding AG 2025).
Xolair treats asthma and allergic conditions and it has generated CHF 3.1 billion in 2025 showing a 32% growth from 2024, with a significant portion coming from the US revenue (Roche Holding AG 2025).
Combined, these three products represent CHF 14.2 billion in revenue, 23% of total 2025 group sales of CHF 61.5 billion (Roche Holding AG 2025). Analysts estimate that full MFN enforcement could cut Roche's US revenue by 15% to 30%, concentrated in single source drugs (Frappé et al. 2025). A 20% reduction across this product base implies an annual revenue loss of approximately CHF 1.9 billion.
The financial impact depends on the administration's chosen strategy. The scenarios are constructed based on the range shown by Frappé et al., 2025. If a more favourable decision is taken, the December 2025 framework remains in place and MFN stays limited to Medicaid, and the annual impact stays below CHF 200 million or less if considering the numbers presented by Public Eye. If a moderate decision is taken, MFN extends to single source drugs in Medicare Part B with an average price reduction near 15%, producing an annual impact of CHF 1.0 to 1.5 billion. The full enforcement of the most extreme decision could potentially produce a much higher annual impact. The moderate range is in line with the sensitivity assumptions used in the financial model explained later in the text.
The MFN reference basket draws from OECD countries with a GDP per capita of at least 60% of the US level (Frappé et al. 2025). Switzerland is included as Swiss drug prices already sit below US levels. Roche's Swiss home market pricing therefore anchors the benchmark that constrains its US revenue. This feedback loop is built into the policy design.
Roche spent CHF 12.24 billion in R&D in 2025, equal to 20% of total sales (Roche Holding AG 2025), but US margin compression would be eating into drug development returns (Roche has 66 molecules in development, with 10 entering Phase III (Roche Holding AG 2025)). The viability of that pipeline depends on the US market remaining commercially attractive for premium priced biologics.

4.3 Supply Chain and other geopolitical risk

Roche currently operates across more than 100 countries with 13 US manufacturing sites and several regional production hubs (Roche Holding AG 2025). This footprint provides a good degree of redundancy however, three specific vulnerabilities still matter.
China supplies approximately 40% of global active pharmaceutical ingredient (API) production (Swissinfo 2026). Roche has cited reducing its Chinese API dependency as a reason for its US investment (Roche Holding AG 2025), which confirms that the exposure is material. The near-term risk is a cost escalation and weaker procurement leverage if trade conditions between the US and China tighten. If Chinese API costs rise it can be expected that the effect on Roche's production cost for affected products is direct. This, however, is a rebalancing problem through several years rather than being an immediate disruption.
Three US distributors account for approximately one third of Roche's total group sales (Roche Holding AG 2025). This is a domestic structural risk with no direct geopolitical cause. Indeed any dispute with a contractor, a distributor acquisition, or a regulatory action against one of these companies would hit revenue immediately. This risk, while relevant, doesn't appear in any published Roche mitigation strategy.
Ocrevus, Vabysmo, and similar treatments often require strict temperature control throughout the supply chain. Red Sea shipping disruptions since late 2024 have increased freight premiums and reduced routing options. Marine insurance premiums on affected routes roughly doubled between late 2024 and early 2026 (Think Global Health, 2026). Re-routing temperature sensitive biologics is not always feasible without risking product integrity. Roche has invested in digital supply chain visibility tools and strategic inventory reserves (Roche Holding AG 2025) (Sustainability Magazine, 2026). The adequacy of these measures against a sustained closure has not been publicly quantified.
This risk affects logistics cost and delivery timing but does not threaten manufacturing capacity, which is geographically distributed. This means that the supply chain risk remains moderate but the distributor concentration is a specific gap that would probably need some attention.

4.4 Risk Prioritization and Strategic Implication

Table 1: Political Risk Summary

Risk	Probability	Potential Impact	Priority
MFN drug pricing	High	High (CHF 1.0 to 1.5 bn annually)	1
Supply chain concentration	Medium	Medium	2
US tariff exposure	Low	High (CHF 1.3 to 3.2bn EBIT)	3

MFN pricing ranks first. The policy is active, the exposed products are named, and full enforcement would generate very significant annual losses . Roche's ability to mitigate depends on negotiation and launch sequencing, neither of which it fully controls.
Supply chain concentration ranks second. The distributor concentration scores lower on magnitude than tariffs, but it is the least addressed risk in Roche's current strategy.
Tariff exposure ranks third. The USD 50 billion investment plan has bought political cover. The remaining risk concerns execution of that commitment.
The MFN risk drives the market exit analysis presented in section 6. Exiting low price reference markets such as Bulgaria reduces the external benchmark used to set MFN targets. Launch sequencing, entering high-price markets first and delaying entry into low-price ones, addresses the same mechanism with less reputational cost.
The tariff risk and the API supply chain risk jointly motivate the production relocation analysis in section 5. Moving selected mature biologics manufacturing to a contract manufacturer in the EU lowers cost exposure on declining products and builds a supply route outside the dependency on the US and China.

5 Production Relocation

Building on the analysis of Roche's strong dependence on the United States and the increasing political and regulatory risks mentioned in the previous sections, production relocation emerges as a key strategic lever. Roche faces a trade-off between localizing production in the United States to mitigate tariff exposure and reducing costs to counter margin pressure resulting from pricing regulation, biosimilar competition, and increasing R&D intensity.
Therefore, production relocation should not be seen as a full shift of production, but rather as a selective extension of the global production for non-core and cost-sensitive manufacturing processes.

5.1 Product Selection for Offshoring

A differentiated and detailed analysis is needed to determine which parts of Roche's portfolio would be suitable for relocation. Pharmaceutical products differ greatly regarding technological complexity, regulatory requirements, and strategic importance.
Highly complex biologics, such as Ocrevus and Vabysmo, represent advanced therapeutic antibodies targeting specific disease mechanisms. Ocrevus is used for multiple sclerosis treatment and requires highly controlled and consistent production processes, this ensures safety and efficacy (Food and Administration 2023a). Vabysmo is used for retinal diseases and incorporates high molecular and process complexity due to its dual-binding mechanism (Food and Administration 2023b).
From a manufacturing standpoint, these next-generation biologics involve complex cell culture processes, highly sensitive purification steps, and strict control of process parameters such as temperature, pH, and protein folding. Even small variations could lead to deterioration of product quality, immunogenicity, and clinical outcomes. In addition, the regulatory requirements for biologics are particularly strict, as even small changes in the production process require large comparability studies and regulatory re-approval (Food and Administration 2022).
As a result, these kinds of products are part of Roche's core capabilities in research, development, and high-quality manufacturing and are typically produced in highly controlled environments, which are close to major markets or target markets. Offshoring such products would introduce significant operational, regulatory, and reputational risks and is hence not recommended.
In contrast, mature biologics such as Avastin, Herceptin, and MabThera are drugs that have been on the market for many years. Their production processes are well-established, extensively documented, and standardized across multiple production sites. In addition, these products have lost patent protection and also face increasing biosimilar competition, which leads to price pressure and decreasing revenues. Roche reports declines of approximately –15% (Avastin), –20% (Herceptin), and –16% (MabThera) respectively (Roche 2025).
While these products are biologics as well and therefore still require careful production control, the associated manufacturing risks are lower compared to newer therapies. The processes are quantifiable, outcomes are more predictable, and technology transfer to external production sites is more feasible.
As a consequence, the relocation strategy focuses on mature biologics and standardized downstream processes (e.g. fill-finish and packaging) for non-US markets, where cost efficiency becomes more important than keeping full internal control.

5.2 Make-or-Buy Decision Using the IfM Framework

Applying the IfM make-or-buy decision framework (Institute for Manufacturing 2007) provides a structured approach to determine the appropriate sourcing strategy. Rather than seeing the decision as a binary choice, the framework guides the analysis through a series of key questions related to strategic importance, knowledge protection, supplier risk, and operational characteristics.
For the selected product category of mature biologics, several key questions of the IfM flowchart are particularly relevant. First, the question of whether critical technology must be protected remains partially relevant. Although these products are no longer at the technological top, manufacturing still relies on

specialized process knowledge and expertise, which means that complete outsourcing could lead to knowledge leakage.
Second, the framework considers whether performing the activity in-house enhances the firm's ability to innovate. In Roche's case with mature biologics, this effect is limited, as these products have been on the market for several years and have also lost patent protection; therefore, they do not contribute significantly to Roche's innovation pipeline. This is an argument against full in-house production.
Third, supplier-related risks must be evaluated. Fully outsourcing production would increase dependency on external partners and could lead to reduced negotiation power and potential vulnerabilities in the supply chain. Considering the limited number of qualified biologics manufacturers, this risk is to be taken seriously.
Finally, the transferability of the work package is a critical factor. Downstream tasks such as fill-finish and packaging are relatively measurable, making them suitable for outsourcing. In contrast, upstream biologics production remains more complex and less easy to transfer.
All in all, the IfM decision logic does not lead to a clear "make" or "buy" outcome, but it points to a hybrid solution. Roche should keep strategically critical products and processes in-house but outsource standardized and transferable production activities.
This corresponds to the IfM "make some" strategy, which allows Roche to balance cost efficiency with strategic control and reduce exposure to supplier risk. In the context of this analysis, this approach is particularly suitable for mature biologics such as Avastin, Herceptin, and MabThera, where production processes are standardized enough to allow partial outsourcing while still having internal control over quality and regulatory compliance.

5.3 Country Short-List

Three countries are selected for further analysis: India, Poland, and Bulgaria.
India is a global pharmaceutical manufacturing hub and has exports of approximately USD 30.5 billion and strong cost advantages (India 2025). However, regulatory variability is a concern, as shown by Reuters (2024).
Poland offers a strong balance between efficiency and regulatory stability. Pharmaceutical exports reached approximately €4.8 billion in 2023 (Investment and Agency 2025). The country benefits from alignment with EU regulations and a growing life sciences ecosystem.
Bulgaria has very low labour costs (approximately €12/hour) but has a relatively small pharmaceutical ecosystem (Eurostat 2026b).

5.4 Weighted Score Analysis

To select the most suitable country, a weighted score model is used following the method introduced in the lecture. The model consists of three steps: (1) identifying relevant criteria, (2) assigning weights, and (3) scoring each country. The total score is calculated as the sum of the product of weights and scores.
The sub-scores are based on a combination of quantitative indicators (e.g., export volumes, labour costs, and logistics conditions) and qualitative assessments (e.g. regulatory stability and ecosystem maturity) as shown in weighted score methodology.

5.4.1 Step 1: Selection Criteria

The following criteria are used: cost competitiveness, regulatory environment and quality standards, geopolitical risk, logistics and infrastructure, pharmaceutical ecosystem and supplier base.
These criteria were selected because they reflect Roche's strategic goals and the operational risks associated with the production of biologics. Cost competitiveness is relevant due to increasing margin pressure from biosimilar competition and political pricing regulation. Regulatory environment is particularly important in the pharma industry where compliance with FDA and EMA standards is non-negotiable in order to have market access. Geopolitical risk is included because the relocation strategy aims to establish

a long-term, low-risk manufacturing solution that is less exposed to political disruptions in the United States. Logistics and infrastructure influence the supply reliability while ecosystem depth reflects how many qualified suppliers and skilled labour there is for biologics manufacturing.
More specifically, cost competitiveness is approximated using relative labour costs and GDP per capita, the regulatory environment is assessed based on compliance with international standards (e.g. FDA/EMA approvals), geopolitical risk is assessed using governance indicators such as political stability and corruption indices, logistics is evaluated using infrastructure quality and supply chain connectivity, and ecosystem depth reflects the size and maturity of the pharmaceutical manufacturing base.
These indicators are based on publicly available international datasets, including GDP per capita and labour cost statistics (Bank 2024; Eurostat 2026b), logistics performance indicators (Bank 2023), and governance measures such as the Corruption Perceptions Index (International 2024).

5.4.2 Step 2: Weighting

Weights are assigned on a scale from 1 (low importance) to 5 (high importance):

Criterion	Weight
Cost competitiveness	5
Regulatory environment	5
Geopolitical risk	4
Logistics	3
Ecosystem depth	3

Table 2: Weights assigned to selection criteria.

The weights above are based on Roche's strategic priorities regarding pharmaceutical manufacturing. Cost competitiveness and regulatory receive the highest importance score, as the primary relocation reason is the cost reduction while the regulatory environment and product quality are not compromised. Geopolitical risk receives a slightly lower score as political stability might or might not influence the manufacturing process and supply chain. Logistics and ecosystem depth receive moderate weights as they remain important for scalability of production but have less direct impact than cost competitiveness or regulatory environment.

5.4.3 Step 3: Scoring

Each country is scored on a scale from 0 to 100:

Criterion	Weight	India	Poland	Bulgaria
Cost competitiveness	5	90	65	85
Regulatory environment	5	60	85	70
Geopolitical risk	4	60	80	60
Logistics	3	70	85	70
Ecosystem depth	3	90	80	50
Total Score		**1470**	**1565**	**1375**

Table 3: Weighted score analysis (Total = $\sum$ weight $\times$ score)

The assigned scores reflect the relative strengths and weaknesses between the countries across the selected criteria. India achieves its highest score in cost competitiveness and ecosystem depth due to its large pharmaceutical industry, strong export base, and comparatively low labour costs. However, it performs weaker in regulatory stability and geopolitical risk due to its higher variability in compliance standards and governance indicators. Poland scores high in many categories, particularly in regulatory environment, logistics, and geopolitical stability, benefiting from the European Union standards, strong infrastructure,

and a growing pharmaceutical ecosystem. Bulgaria performs well in cost competitiveness but lacks ecosystem depth and geopolitical stability, therefore achieving the lowest score across the three countries. It is important to note that these scores reflect relative differences between countries based on the underlying indicators rather than precise quantitative measurements.

5.5 Implementation Strategy

Based on the findings in 5.2 and therefore the "make some" strategy, Roche should pursue a contract manufacturing approach rather than a greenfield investment.
A suitable partner is Rezon Bio, a Polish biologics CDMO with GMP-compliant facilities inspected by regulatory authorities such as the FDA and EMA (Bio 2025).
A step-by-step outsourcing strategy is recommended:

- Initial transfer of limited production volumes for selected mature biologics
- Gradual scale-up after successful quality validation and regulatory approval
- Retention of Roche's core capabilities (process control, quality assurance, and final product release)

6 Strategic market exit

The Most-Favoured-Nation (MFN) principle was established after World War II to reduce trade friction and power imbalances between countries. Under MFN rules, any favourable trading conditions granted to one country must automatically be extended to all other trading partners. This principle, central to the World Trade Organization (WTO), promotes fairness and equality in international trade and supports a predictable environment for long-term business planning and investment. (WTO n.d.)
Recently, Donald Trump proposed applying the MFN principle to reduce drug prices in the United States, arguing that other developed countries have benefited from relatively low pharmaceutical prices, while high US prices have financed global innovation. Under this proposal, pharmaceutical companies would be required to offer US consumers the lowest available price among comparable countries, a policy with significant strategic implications for Roche, which operates extensively in the United States. (House n.d.; Roche n.d.[a])

6.1 Should the company strategically exit any specific market in response?

One possible strategic response for Roche would be to exit markets that generate relatively low revenues and have low drug prices. By doing so, the reference price used under MFN could increase, allowing Roche to maintain higher prices and profitability in the US. Given its scale and pricing levels, leaving the US itself is not a viable option. As seen in the graphic, the US accounts for approximately 53% of pharmaceutical sales. Instead, smaller markets with lower prices and limited revenue potential become candidates. (Roche 2026)

Pharmaceuticals Division – Sales by region

	2025 (CHF m)	2024 (CHF m)	% change (CER)	% of sales 2025	% of sales 2024
United States	25,355	24,774	+8	53.2	53.7
Europe	9,164	8,832	+5	19.2	19.1
Japan	2,882	2,874	+5	6.0	6.2
International	10,268	9,691	+14	21.6	21.0
– *of which China*	*3,034*	*2,926*	*+10*	*6.4*	*6.3*
Total sales	**47,669**	**46,171**	**+9**	**100**	**100**

Figure 5: Roche (2026)

However, such a strategy conflicts with Roche's sustainability goal of "Access to Innovation", which aims to ensure that patients around the world have access to essential medicines. This creates a tension between profit maximization and global health equity. (Roche n.d.[b])
The MFN pricing model proposed by Trump is based on a basket of countries, of the Organization for Economic Co-operation and Development (OECD), with a GDP per capita of at least 60% of that of the US. This basket includes countries such as Australia, Austria, Belgium, Canada, Germany, Denmark, Finland, Iceland, Ireland, Israel, Luxembourg, Netherlands, Norway, Sweden, Switzerland, and the United Kingdom. When adjusted for purchasing power parity, additional countries such as France, Italy, Spain, Czech Republic, Japan, Lithuania, New Zealand, Slovenia, and South Korea are also included. (Frappé et al. 2025)

For Roche leaving a country would only be beneficial if:

- The country generates relatively low revenue
- Drug prices are significantly lower than in other reference countries
- The country influences the pricing in other markets that are references or is itself a reference country.

The last point matters considerably, as many countries use international reference pricing themselves. Slovakia, for example, influences prices in countries like Austria and Germany, Therefore, leaving Slovakia could indirectly increase prices in multiple markets, including the US. (Frappé et al. 2025)

Low price countries such as Bulgaria, Croatia and Latvia are frequently used as reference markets while contributing relatively little to Roche's overall revenue. Drug prices in Croatia have been found to be up to eleven times lower than in Germany for certain drugs, and Bulgaria is considered one of the lowest priced pharmaceutical markets in Europe (Zaprutko et al. 2017). Bulgaria's strict reference pricing system, benchmarked against a basket of twelve countries, creates a continuous downward pressure on prices and makes it a strong candidate for strategic exit (Labban n.d.).
From a strategic perspective, Roche could consider maintaining manufacturing operations in Bulgaria due to its cost advantages, while withdrawing from commercial sales. This would avoid the pricing spillover effects while preserving operational efficiency. Importantly, Bulgaria also influences pricing in other European countries such as Germany, which is itself part of the US MFN reference basket, making its commercial presence therefore a risk to higher-value markets. (Labban n.d.)
Beyond market exit, launch sequencing may be an even more powerful lever. By launching first in high-price markets and deliberately delaying entry into lower-price markets, Roche can front-load profits and protect returns while still eventually reaching global patients. This avoids the reputation risk of permanent market withdrawal while still mitigating the downward pressure on reference prices. (Consulting n.d.)

6.2 Should the company strategically enter any specific market in response?

An important strategic complementary response is to strengthen the presence in markets not included in international reference pricing systems, particularly those that do not directly influence US drug price benchmarks. China, Brazil, Saudi Arabia, and the UAE are especially attractive on this ground. Roche already has facilities in all four regions and China is already the largest contributor within the international sales area for pharmaceuticals. (Roche n.d.[a]; Roche 2026)
Expanding in these regions allows Roche to grow revenues without lowering global price benchmarks, diversify geographic risk away from the US and Europe, and maintain greater pricing flexibility than in tightly regulated European markets.

Diagnostics Division – Sales by region

	2025 (CHF m)	2024 (CHF m)	% change (CER)	% of sales 2025	% of sales 2024
Europe, Middle East and Africa (EMEA)	4,965	4,822	+6	35.9	33.7
North America	4,444	4,335	+9	32.1	30.3
– *of which US*	*3,900*	*3,852*	*+7*	*28.2*	*26.9*
Asia-Pacific	3,386	4,099	−12	24.4	28.6
– *of which China*	*1,726*	*2,402*	*-24*	*12.5*	*16.8*
Latin America	1,052	1,068	+11	7.6	7.4
Total sales	**13,847**	**14,324**	**+2**	**100**	**100**

Figure 6: Roche (2026)

China stands out as a particularly strategic country given its large population, expanding middle class, and fast-growing healthcare spending. The China pharmaceutical market is projected to grow at an annual growth rate of 5.47% between 2026 and 2030, reaching a market volume of USD 164.62 billion by 2030. Although the Diagnostics Division experienced a 6% decrease due to the reform of healthcare pricing, the Pharmaceutical Division still achieved a 10% growth, suggesting that a focused strategy can produce strong results. A promising approach would be to negotiate more favourable market conditions in exchange for expanding R&D activities, production capacity, and the presence of a stronger sales force in China. (Roche n.d.[a]; Statista n.d.; Roche 2026)
Brazil offers significant long-term potential driven by a rapidly aging population. Currently around 11% of the population is 65 or older, by 2040 this number is expected to double. The Brazilian pharmaceutical market is expected to grow from USD 41 billion in 2025 to USD 74 billion by 2032. Brazil's growing prevalence of chronic diseases, particularly cancer and diabetes, further strengthens this opportunity, as both fall within Roche's core areas of expertise. (ThinkGlobalHealth n.d.; FortuneBusinessInsigths 2026)
Saudi Arabia and the UAE present strong growth opportunities due to high national wealth and substantial government investment in healthcare infrastructure, supporting higher per-capita healthcare spending and greater willingness to adopt premium-priced treatments. Their limited role in international reference

pricing further reduces the risk of impacting global price benchmarks. The Saudi Arabia pharmaceutical market is projected to grow from USD 10.8 billion in 2025 to USD 16.52 billion by 2032 and the UAE market from USD 4.45 billion in 2025 to USD 8.59 billion by 2034. (Tawfik et al. 2021; Ashraf et al. 2025; FortuneBusinessInsights 2026; ImarcGroup 2026)

Since Roche already operates in all four regions, deepening its presence offers clear advantages over entering new markets.

- Lower entry costs: Existing infrastructure, regulatory approvals, and partnerships reduce the need for large upfront investments
- Faster scalability: Roche can expand production, distribution, and sales more quickly
- Better market knowledge: Familiarity with local regulations, pricing systems, and patient needs improves efficiency and competitiveness

The MFN pricing policy presents Roche with a structurally difficult environment in which profitability, global access commitments, and innovation incentives must be balanced simultaneously. Several tactical responses are available, leaving low-revenue markets like Bulgaria, sequencing launches to front-load returns, and deepening presence in non-reference markets. Each offers a degree of protection, but none resolves the underlying tension cleanly.
That tension runs deeper than pricing mechanics along the way. Roche's public commitment to global access sits uncomfortably against strategies that delay or deny market entry, however financially rational. More broadly, if MFN pricing significantly compresses US revenues, the financial case for high-risk drug development weakens across the entire industry, with consequences for future innovation that extend well beyond Roche. (Slabodkin n.d.)

7 Financial Assessment

7.1 Assumptions

Table 4: Core assumptions used in the financial model

Assumption	Base value	Rationale
Discount rate / WACC	8.0%	Reasonable large-pharma hurdle rate for operational projects with regulatory risk.
Planning horizon for core comparison	5 years	Matches a medium-term case-study decision horizon and captures ramp-up.
Strategic horizon for growth-market option	10 years + terminal value	Market-entry benefits are back-loaded and cannot be judged fairly on a 5-year horizon only.
Roche group sales anchor	CHF 61.5bn	Used to scale exposures and avoid unrealistic option sizes.
Mature biologics sales pool	CHF 3.25bn	Approximation for Avastin, Herceptin and MabThera after decline.
Offshorable sales equivalent	40% of mature biologics sales	Only non-core, standardized and non-US downstream volume is considered.
COGS ratio on mature biologics	25% of sales	Assumes high gross margin but increased price pressure from biosimilars.
Downstream/manufacturing portion of COGS	30% of COGS	Only fill-finish, packaging and selected downstream activities are transferred.
Poland CDMO net cost saving	18% of transferred downstream cost	Lower labour and operating cost, partly offset by CDMO margin and quality oversight.
Annual Roche oversight cost	CHF 8m	Quality assurance, audits, regulatory maintenance, supplier management.
Initial transfer/validation cost	CHF 60m	Technology transfer, comparability studies, validation batches, contracts and dual-running.

The financial values are scenario estimates rather than forecasts. They are calibrated using Roche's reported sales base from the year 2025, industry COGS benchmarks, pharmaceutical quality-risk frameworks, and policy/regulatory sources.

7.2 Option A – Financial analysis of production relocation to Poland

7.2.1 Scope of the option

The production relocation option is not a full relocation of Roche's biologics manufacturing. It is a controlled "make some" model: Roche keeps process ownership, quality release, regulatory responsibility and strategically critical upstream biologics capabilities in-house, while outsourcing selected standardized downstream and packaging steps for mature biologics to a qualified Polish CDMO. This matches the logic of the IfM make-or-buy framework: activities that are highly strategic, knowledge-sensitive or innovation-relevant remain internal, while measurable, mature and transferable work packages can be outsourced under strong governance (University of Cambridge Institute for Manufacturing 2001; Canez, Platts, and Probert 2000).

The relevant financial question is therefore not whether Roche should move a large share of its manufacturing base abroad. The question is whether transferring a limited volume of mature biologics downstream work to Poland creates enough cost saving and risk mitigation to justify the transfer cost and execution risk.

7.2.2 Direct cash-flow model

The transferred sales equivalent is assumed to be 40% of the mature biologics sales pool, equal to CHF 1.30bn. Applying a 25% COGS ratio gives CHF 325m of production-related cost. Since only downstream/fill-finish/packaging is transferred, 30% of COGS is considered addressable, equal to CHF 97.5m. An 18% net cost advantage in Poland yields gross annual savings of CHF 17.55m at full ramp-up. After Roche's annual oversight cost of CHF 8m, the recurring net saving is only CHF 9.55m. This illustrates a central finding: pharmaceutical offshoring has much lower apparent savings than simple labour-arbitrage reasoning suggests, because quality, compliance and dual-control costs are substantial.

Table 5: Option A direct cash-flow model (CHFm)

Year	Ramp-up	Gross saving	Oversight cost	Net cash flow
0	Implementation	-	-	-60.00
1	30%	5.27	-8.00	-2.74
2	60%	10.53	-8.00	2.53
3	100%	17.55	-8.00	9.55
4	100%	17.55	-8.00	9.55
5	100%	17.55	-8.00	9.55

At an 8% discount rate, the five-year NPV of the direct-cost case is approximately CHF -39.3m. The direct payback period is beyond five years. On direct costs alone, the project should not be approved as a conventional cost-cutting initiative.

7.2.3 Including resilience and tariff-risk option value

However, the direct model understates the value of the option. The case context is political risk: US tariffs, MFN pricing, geopolitical volatility and pressure on margins. A validated EU manufacturing partner gives Roche a credible alternative production route for mature products. This does not remove the US exposure of Roche's most important growth drugs, but it reduces the cost base of declining products and provides supply-chain flexibility. To quantify this, the model adds an annual resilience value of CHF 20m from Year 2 onward. This value is not a saving in normal operations; it is the expected value of avoided disruption, avoided emergency sourcing, improved bargaining power with internal sites and external suppliers, and faster reallocation under tariff or reference-pricing pressure.

Table 6: Option A risk-adjusted cash-flow model (CHFm)

Year	Direct net cash flow	Added resilience value	Risk-adjusted cash flow
0	-60.00	0.00	-60.00
1	-2.74	0.00	-2.74
2	2.53	20.00	22.53
3	9.55	20.00	29.55
4	9.55	20.00	29.55
5	9.55	20.00	29.55

With this risk-adjusted value included, the NPV becomes approximately CHF +22.1 m. The conclusion is therefore conditional: the Polish CDMO option is unattractive if it is sold internally as pure cost reduction, but attractive if framed as a resilience investment with a limited transfer scope. The business case should therefore be approved only as a pilot with stage gates: transfer a small volume first, validate quality and regulatory performance, then scale only after evidence of stable cost and service performance.

7.2.4 Sensitivity of Option A

Table 7: Option A sensitivity analysis

Scenario	Key change	Approx. NPV	Interpretation
Pessimistic	Cost saving only 10%; no resilience value	CHF -57m	Do not scale. Only maintain as learning pilot if strategically necessary.
Base direct	18% saving; no resilience value	CHF -39m	Financially weak as a cost programme.
Base risk-adjusted	18% saving + CHF 20m annual option value from Year 2	CHF +22m	Approve as a controlled resilience investment.
Upside	25% saving + CHF 30m annual option value	CHF +66m	Scale further if quality and regulatory KPIs are met.

The sensitivity confirms that the decisive driver is not the hourly labour-cost difference. It is the strategic value of having a qualified, flexible, EU-aligned production alternative. The recommendation is to pursue the option, but to keep it narrow: mature biologics, downstream steps, non-US volume first, and strong Roche quality control.

7.3 Option B – Financial analysis of strategic exit from Bulgaria

7.3.1 Logic of the option

The market-exit option is motivated by the MFN scenario. The 2025 US MFN executive order directed the US administration to communicate MFN price targets to pharmaceutical manufacturers and, if necessary, pursue rulemaking to impose MFN pricing (The White House 2025). CMS subsequently described the GENEROUS Medicaid model as a voluntary five-year model designed to align Medicaid net prices with what selected other countries pay (Centers for Medicare & Medicaid Services 2026). In this policy context, very low-price European markets can create price-pressure risk that is larger than their local revenue contribution.

The earlier report identifies Bulgaria as a potential candidate because it is a low-price market, has strict reference-pricing rules and limited commercial scale for Roche. A financial model must therefore compare two effects: lost Bulgarian operating profit versus the value of preventing Bulgaria from depressing prices in larger reference chains.

7.3.2 Base-case cash-flow model

The base model assumes that Bulgaria contributes only 0.05% of Roche group sales, or roughly CHF 31m annually. At a 25% local operating margin, the lost operating profit from exiting is about CHF 7.8m per year. The model also assumes CHF 12m one-off exit costs for legal work, supply obligations, inventory, stakeholder management and patient-access mitigation. The benefit comes from an assumed CHF 33m annual contribution effect if the exit prevents a small but meaningful downward reference-price effect on higher-value markets, especially the US. Net annual benefit is therefore approximately CHF 20m.

Using a simplified net recurring benefit of CHF 20m and one-off exit cost of CHF 12m, the five-year NPV is approximately CHF +67.9m at an 8% discount rate. On a narrow financial basis, Bulgaria exit looks more attractive than production relocation.

7.3.3 Why the result is fragile

The attractiveness of the exit option depends almost entirely on a legal-pricing mechanism that Roche may not fully control. If Bulgarian prices remain indirectly visible through parallel trade, historical

Table 8: Option B base-case annual effects

Component	Assumption	Annual CHFm effect
Lost Bulgaria revenue	0.05% of CHF 61.5bn group sales	-30.8 revenue
Lost Bulgaria operating profit	25% local operating margin on lost revenue	-7.8 EBIT
Reference-pricing protection	Small uplift/protection on larger US/EU revenue base	+33.0 contribution
Ongoing access/legal/reputation management	Patient continuity and stakeholder management	-5.0 cost
Net recurring effect	Reference benefit minus lost profit and mitigation cost	+20.2 cash flow

reference periods, EU cross-country data, therapeutic substitutes or payer negotiation benchmarks, then exiting the market may not materially raise the reference price. In that case, Roche loses local revenue and reputation while gaining little. The financial NPV would quickly turn negative. There is also a non-financial cost: market withdrawal conflicts with Roche's public access-to-medicine commitments and may trigger political scrutiny in Europe. Therefore, this option should not be treated as a broad exit strategy. It is a tactical lever to be used only if legal review confirms that withdrawal would actually remove the relevant price from the MFN or reference-pricing basket.

Table 9: Option B sensitivity analysis

Scenario	Reference-price effect achieved?	Approx. NPV	Decision
No effect	0%	CHF -43m	Do not exit; maintain supply and renegotiate pricing.
Partial effect	40% of assumed benefit	CHF +1m	Not attractive after reputational risk.
Base case	100% of assumed benefit	CHF +68m	Financially attractive but politically sensitive.
Strong effect	150% of assumed benefit	CHF +124m	Financially strong; still requires access mitigation.

The recommendation is therefore "do not execute immediately." Roche should prepare a market-exit playbook, but first explore less disruptive alternatives such as launch sequencing, confidential managed-entry agreements, indication-specific pricing, patient access programmes, or reducing the set of products sold at externally visible low prices.

7.4 Option C – Financial analysis of market deepening in China, Brazil, Saudi Arabia and the UAE

7.4.1 Logic of the option

Task 5b argues for strengthening Roche's presence in markets that are attractive for growth but do not directly lower the US MFN benchmark. The relevant markets are China, Brazil, Saudi Arabia and the UAE. The option is not a classic greenfield market entry because Roche already has a global presence; it is better understood as market deepening through distribution expansion, local partnerships, selected R&D localization, medical affairs, market access capabilities and product-launch prioritization.
Financially, this option differs from Bulgaria exit and Poland outsourcing. It requires meaningful upfront spending and its payoff is slower. A five-year view penalizes it because market access, reimbursement and physician adoption take time. A ten-year view is more appropriate because the option builds a growth

platform and diversifies Roche away from the US/EU reference-pricing axis.

7.4.2 Investment and cash-flow model

The base case assumes CHF 250m in cumulative investment over the first three years: CHF 100m in Year 0, CHF 50m in Year 1 and CHF 100m in Year 2. Incremental revenue ramps from CHF 100m in Year 1 to CHF 650m by Year 5. A 22% contribution margin is applied after local pricing, distribution, medical affairs and market-access costs. Additional local operating costs ramp from CHF 30m to CHF 90m. Under this five-year lens, the project remains negative because scale is still building.

Table 10: Option C five-year cash-flow model (CHFm)

Year	Incremental revenue	Contribution at 22%	Extra local OPEX	Additional investment	Net cash flow
0	0	0	0	-100	-100
1	100	22	-30	-50	-58
2	220	48.4	-45	-100	-96.6
3	360	79.2	-60	0	19.2
4	500	110.0	-75	0	35.0
5	650	143.0	-90	0	53.0

The five-year NPV is approximately CHF -159m. This does not mean the option is unattractive; it means the option cannot be justified as a short payback project. Extending the model to Year 10, with net cash flows increasing to CHF 120m by Year 10, gives a positive NPV of approximately CHF +99m before terminal value. If a conservative terminal value is included at 2% perpetual growth after Year 10, the NPV rises above CHF 1.0bn. Even if the terminal value is discounted heavily for political and reimbursement risk, the long-term economics are materially stronger than the five-year view suggests.

7.4.3 Country prioritisation inside the growth-market option

Table 11: Country prioritisation inside the growth-market option

Market	Financial attractiveness	Main upside	Main risk	Suggested priority
China	High long-term scale; medium price pressure	Large patient base, diagnostics-pharma synergies, existing Roche footprint	Centralized procurement and price cuts	Priority 1, but focus on differentiated products
Saudi Arabia	High value per patient; government healthcare investment	Premium innovative medicines and localization policy support	Market access dependency on public procurement	Priority 2
UAE	High willingness to pay; regional hub potential	Premium positioning and fast adoption in specialized care	Small population limits scale	Priority 3, as regional hub
Brazil	Large population; medium margins	Diversification and volume growth	Currency volatility and reimbursement complexity	Priority 4, selective expansion

A disciplined version of the option should therefore not spread investment evenly. Roche should prioritize China for scale but protect margins by focusing on innovative, clinically differentiated products and diagnostics-linked care pathways. Saudi Arabia and the UAE should be used for premium access and regional positioning. Brazil should be approached selectively because its volume potential is attractive but currency and reimbursement risk can erode the contribution margin. Country-level data sources such as the World Bank, Eurostat and Transparency International should be used to validate the macroeconomic and institutional assumptions before final investment approval (World Bank 2026; Eurostat 2026a; Transparency International 2026).

7.5 Cross-option comparison and recommendation

The options should not be ranked only by five-year NPV. If Roche wants immediate financial protection against MFN pricing, Bulgaria exit appears best, but it is legally fragile and strategically controversial. If Roche wants operational resilience, the Poland CDMO pilot is the most balanced option, but it must be justified by risk-adjusted flexibility rather than direct savings. If Roche wants long-term revenue diversification, market deepening is the strongest option, but it requires patience and disciplined investment.

Table 12: Cross-option comparison

Option	5-year NPV	10-year / strategic value	Risk level	Recommended action
A. Poland CDMO production relocation	CHF -39m direct; CHF +22m risk-adjusted	Medium; creates operational flexibility and cost discipline	Medium execution and quality risk	Approve as staged pilot, not as broad relocation.
B. Bulgaria commercial exit	CHF +68m base case	Low to medium; mostly tactical pricing protection	High legal, access and reputation risk	Prepare but do not execute until reference-price effect is legally confirmed.
C. Deepen non-reference growth markets	CHF -159m in 5-year view	High; CHF +99m over 10 years before terminal value, much higher with terminal value	Medium to high market-access risk	Pursue selectively with phased investment and country prioritisation.

The recommended portfolio is therefore:

- Proceed with Option A as a narrow, staged Polish CDMO pilot for mature biologics downstream work. Use hard stage gates: validated batches, regulatory approval, on-time-in-full supply performance, deviation rate, and audited cost per batch.
- Do not immediately execute Option B as a full Bulgaria exit. First commission a legal and pricing-system analysis to confirm whether Bulgarian withdrawal would actually change the relevant MFN or international reference price. If confirmed, consider a partial product-level exit combined with patient-access safeguards.
- Proceed with Option C as a phased growth-market programme, but use a ten-year investment case. Prioritize China, Saudi Arabia and the UAE for value, while using Brazil selectively for scale. Tie investment tranches to reimbursement wins, product launches and contribution-margin milestones.

7.6 Final conclusion

Roche's financial exposure cannot be solved by a single move. Its US dependence is too large, its products are too regulated, and its global reputation depends on access to medicines. The financially strongest narrow move is a Bulgaria exit, but it creates access and political risk and only works if the reference-price mechanism changes as expected. The most operationally robust move is the Polish CDMO pilot, which should be pursued as a resilience investment despite weak direct cost savings. The most strategically valuable long-term move is expansion in non-reference growth markets, because it diversifies future revenue away from the US/EU pricing corridor. A sensible financial recommendation is therefore a portfolio response: use production relocation to protect margins and flexibility, use market-exit logic only tactically, and invest in selected growth markets for long-term diversification.

7.7 Calculation formulas

The main formula used is:

$$\text{NPV} = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t}, \tag{1}$$

where CF_t is the cash flow in year t, r is the discount rate and T is the planning horizon. For production relocation, annual savings equal transferred downstream cost multiplied by the net percentage saving,

minus annual oversight cost. For market exit, annual net benefit equals reference-pricing protection minus lost local operating profit and mitigation cost. For market deepening, annual net cash flow equals incremental revenue times contribution margin, minus incremental operating expenditure and additional investment.

Table 13: Formula summary

Formula item	Expression	Base result
Transferred mature-biologics sales	CHF 3.25bn x 40%	CHF 1.30bn
Addressable downstream cost	CHF 1.30bn x 25% COGS x 30%	CHF 97.5m
Full-run gross saving in Poland	CHF 97.5m x 18%	CHF 17.55m
Full-run direct net saving	CHF 17.55m - CHF 8m oversight	CHF 9.55m
Bulgaria sales exposure	CHF 61.5bn x 0.05%	CHF 30.8m
Bulgaria lost local profit	CHF 30.8m x 25%	CHF 7.7m
Growth-market contribution	Incremental revenue x 22%	Varies by year

8 Conclusion

Roche operates from a position of structural strength, but faces a growing strategic vulnerability. Switzerland's institutional role, which bridges major regulatory markets and fosters a world-class innovation ecosystem, provides Roche with a strong home base. However, Roche's concentration risk is significant: roughly half of its revenues are derived from the US, its single largest market, making it acutely exposed to American policy changes. This exposure is compounded by intensifying global competition, particularly the price pressure emerging from Asia.
The most immediate political threat is the potential implementation of Most-Favored-Nation (MFN) drug pricing in the US, which could significantly compress margins on Roche's highest-revenue products. Supply chain disruption and tariff risk add further uncertainty. That said Roche's financial resilience is notable, even under tariff scenarios, a 31% net profit margin is projected. Switzerland's unique geopolitical position also provides Roche with a negotiation leverage that has been underutilized, so it is recommended to coordinate more closely with the Swiss government to assert this leverage in bilateral trade discussions. To mitigate MFN pricing specifically, we recommend a disciplined launch sequencing strategy, managing the order in which new drugs enter reference and non-reference markets, as the primary tool, rather than market exit, which carries unacceptable reputational and political costs

Two of the analyzed opportunities stand out as priorities:
Offshoring mature biologics production to Poland is the most actionable near-term step. Relocating mature biologics and standardized downstream processes to a Polish contract manufacturer reduces operational risk and builds supply chain resilience, without compromising Roche's core R&D and high-complexity manufacturing capabilities. This should be framed as a resilience investment, not a cost-cutting exercise. Expanding into non-reference growth markets, particularly China, Saudi Arabia, and the UAE, offers the most credible path to long-term revenue diversification. These markets are not MFN reference countries, carry strong growth in healthcare spending, and reduce Roche's structural dependence on the US. China, given its size, deserves particular attention. Roche could negotiate favorable regulatory terms in exchange for deepening its R&D, manufacturing, and commercial presence there. Brazil, while attractive in size, presents significant currency and reimbursement risks and should not be prioritized relative to Saudi Arabia and the UAE.

Beyond the scenarios analyzed, three additional strategies deserve consideration. First, integration of AI across the value chain. Incorporation of AI in drug discovery and clinical development could significantly reduce R&D expenditure, one of Roche's largest cost drivers, while accelerating time-to-market. In diagnostics, AI-enhanced solutions should also reinforce the premium pricing rationale for Roche's products. Second, proactive government engagement, not only with Switzerland but with local governments in target markets, could smooth regulatory pathways and convert political relationships into competitive advantages. Third, Roche should conduct a comprehensive portfolio stress-test against MFN scenarios, identifying which products are most exposed, and ensuring launch sequencing plans are in place before any policy implementation

In conclusion, Roche should not exit any market. Instead, it should pursue a dual strategy: defend its US position through launch sequencing and government engagement while systematically diversifying into growth markets. Combined with targeted production offshoring and AI-driven efficiency gains, this positions Roche to remain resilient across a range of geopolitical and competitive scenarios without compromising its core commitment to innovation access.

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A AI use declaration

We declare that AI was used as an auxiliary tool during the preparation of this report. It was used to improve language, clarity and structure, to support understanding of financial-modelling concepts and to check the plausibility and presentation of selected scenario-based estimations. AI was not used as an authoritative source. All assumptions, calculations, citations, interpretations and final conclusions were reviewed, selected and approved by the authors. The authors take full responsibility for the submitted work.